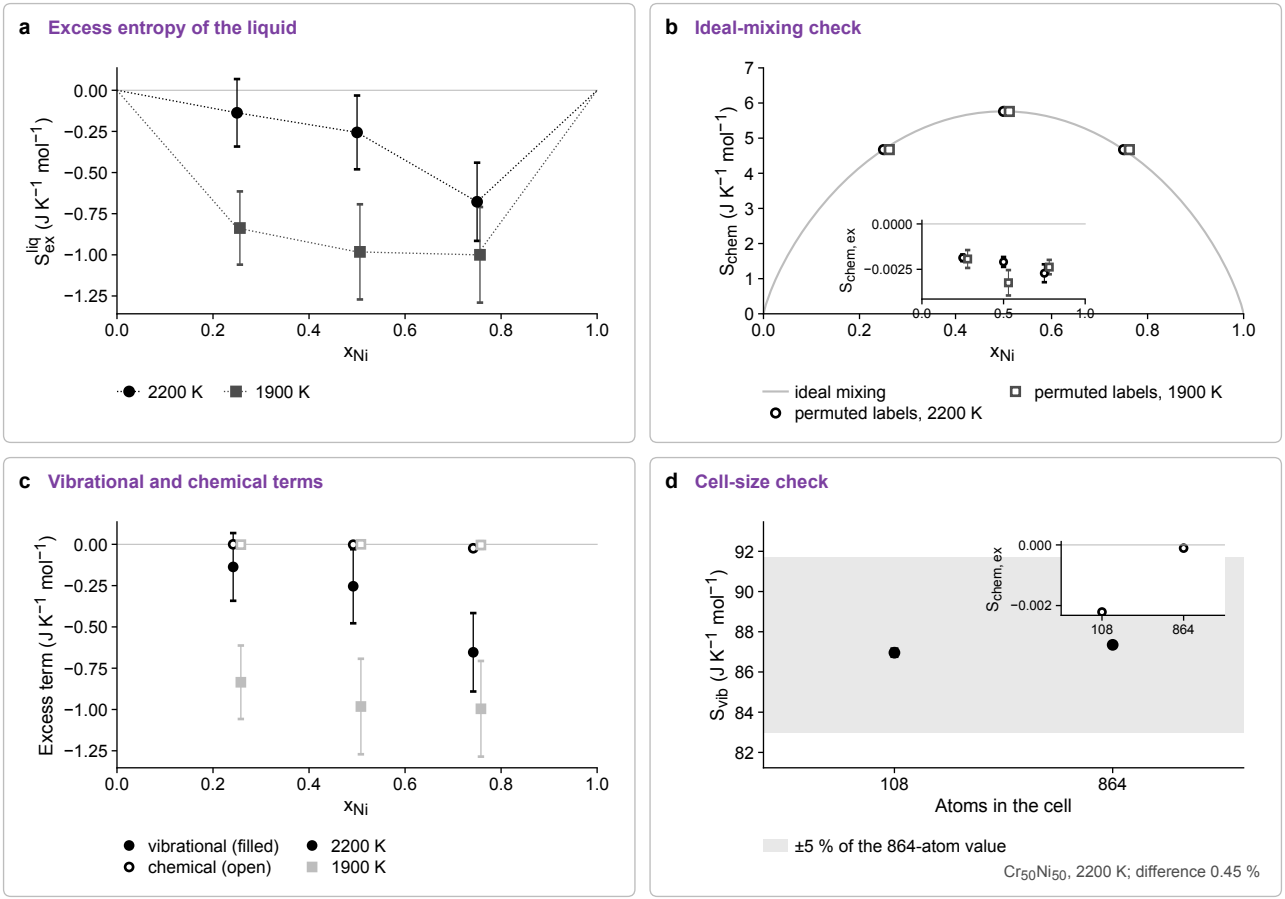
a
Excess entropy of the liquid
S_{ex}^{liq} (J K^{-1} mol^{-1})
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
0.0
0.2
0.4
0.6
0.8
1.0
x_{Ni}
2200 K
1900 K
b
Ideal-mixing check
S_{chem} (J K^{-1} mol^{-1})
7
6
5
4
3
2
1
0
$S_{chem.ex}$
0.0000
−0.0025
0.0
0.5
1.0
x_{Ni}
ideal mixing
permuted labels, 2200 K
permuted labels, 1900 K
c
Vibrational and chemical terms
Excess term (J K^{-1} mol^{-1})
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
0.0
0.2
0.4
0.6
0.8
1.0
x_{Ni}
vibrational (filled)
chemical (open)
2200 K
1900 K
d
Cell-size check
S_{vib} (J K^{-1} mol^{-1})
92
90
88
86
84
82
$S_{chem.ex}$
0.000
−0.002
108
864
108
864
Atoms in the cell
±5 % of the 864-atom value
$Cr_{50}Ni_{50}$, 2200 K; difference 0.45 %